

Likith Kumar Kandukuri

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EDUCATION

University at Albany, SUNY

B.S. in Electrical and Computer Engineering

Albany, NY

Aug. 2025 – May 2029

EXPERIENCE

Undergraduate Research Assistant

May 2025 – Present

Ion Beam Lab, University at Albany – Prof. Andrew Knutson

Albany, NY

- Designed and built a multi-node flow monitoring and safety control system for active ion beam lab equipment, connecting multiple Arduino Nano sensor nodes over RS-485 serial communication on a single line.
- Wrote a Python backend that continuously reads live sensor data, logs it to daily CSV files, and fires automated SMTP email alerts the moment a fault is detected – keeping the lab informed without anyone watching a screen.
- Built and deployed a live Flask web dashboard so lab staff can check real-time flow data from any browser without touching the hardware or the terminal.

Scouting and Strategy Specialist

Mar. 2025

FIRST Robotics Competition – Tech Valley Regional

Albany, NY

- Tracked match and pit data on 40+ teams to guide alliance picks and in-match adjustments for the drive team.

PROJECTS

Cloud Krida | React, Vite, Supabase, PostgreSQL, Vercel | cloudkrida.com

May 2026 - Present

- Built and deployed a full-stack tournament management platform used by real community organizations; hosted the Albany Telangana Community Chess Tournament 2026 across 4 age divisions with 100+ participants.
- Implemented Swiss Points and Single Elimination bracket systems with live pairings, board assignments, match history, and per-tournament photo galleries.
- Used Supabase for PostgreSQL database, authentication, file storage, and role-based access control across guest, admin, and super admin tiers; deployed on Vercel with a custom domain.

Tunnel Vision | Godot, Python, OpenCV, MediaPipe, WebSockets | 4th Place – Hack-A-Damien

Apr. 2026

- Built the entire hand-tracking pipeline for a 5-person team's hackathon game, using OpenCV and MediaPipe to detect and classify gestures from a live webcam feed – placed 4th out of 25 teams in 24 hours.
- Streamed gesture events from Python into Godot in real time over WebSockets, with Flask managing the data flow between the tracking backend and the game engine.
- Owned the most technically unfamiliar component of the stack solo, enabling the rest of the team to focus on game development and art under a one-input-only constraint.

Line Follower Robot | Arduino, IR Sensors | Most Creative Build, UAlbany Showcase

Apr. 2026

- Won Most Creative Build out of 20 teams at the UAlbany Showcase 2026; designed the robot's sensor layout and light-interference mitigation strategy to ensure reliable IR detection under varying ambient light conditions.
- Engineered a priority-based steering cascade where outer sensors trigger sharp turns and inner sensors handle fine corrections; tuned pulse durations per case to achieve smooth, accurate tracking.

ACTIVITIES

Officer

Aug. 2025 – Present

IEEE Student Branch, University at Albany

Albany, NY

- Evaluated potential sponsors and advised on outreach strategy, helping identify companies aligned with the branch's technical focus areas.
- Delivered structured post-event feedback on Hack-A-Damien, proposing introductory workshops to lower the barrier for first-time participants.
- Contributed to branch direction discussions across multiple semesters, offering input on event format, structure, and member engagement.

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, HTML/CSS

Hardware & Embedded: Arduino, RS-485, IR Sensors, Relay Modules, Motor Control

Tools & Frameworks: Git, VS Code, OpenCV, MediaPipe, Flask, PyCharm, IntelliJ